

Metabolism_Unified

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Overview

This code cleans up the metabolism data, ran on a few different days.

The raw data (and code for each day ran) is saved in a subfolder where this code is saved.

Set Up Environment

Load and Clean Data

Flow Data

Instantaneous

Find SD for calendar year at each site

Metabolism

Ellerbe Creek

New Hope Creek

Merge Together and Clean

One question - should we replace unrealistic values (neg GPP, pos ER) with zero (as done below), or remove them all together from the dataset (equivalent to a day when the model didn't converge)?

Remove high Q days

High flow is defined as being a discharge value that is greater than 1 standard deviations above that site's average discharge.

we will flag days that exceed 1 SD.

Not many of the days, but definitely a few

Days_Modelled	forested	stormwater dominated	wastewater dominated
Exceeds Discharge Threshold	13	21	42
GPP/ER Unrealistic	34	86	29
No Model Output	136	110	23
Passed	211	177	300

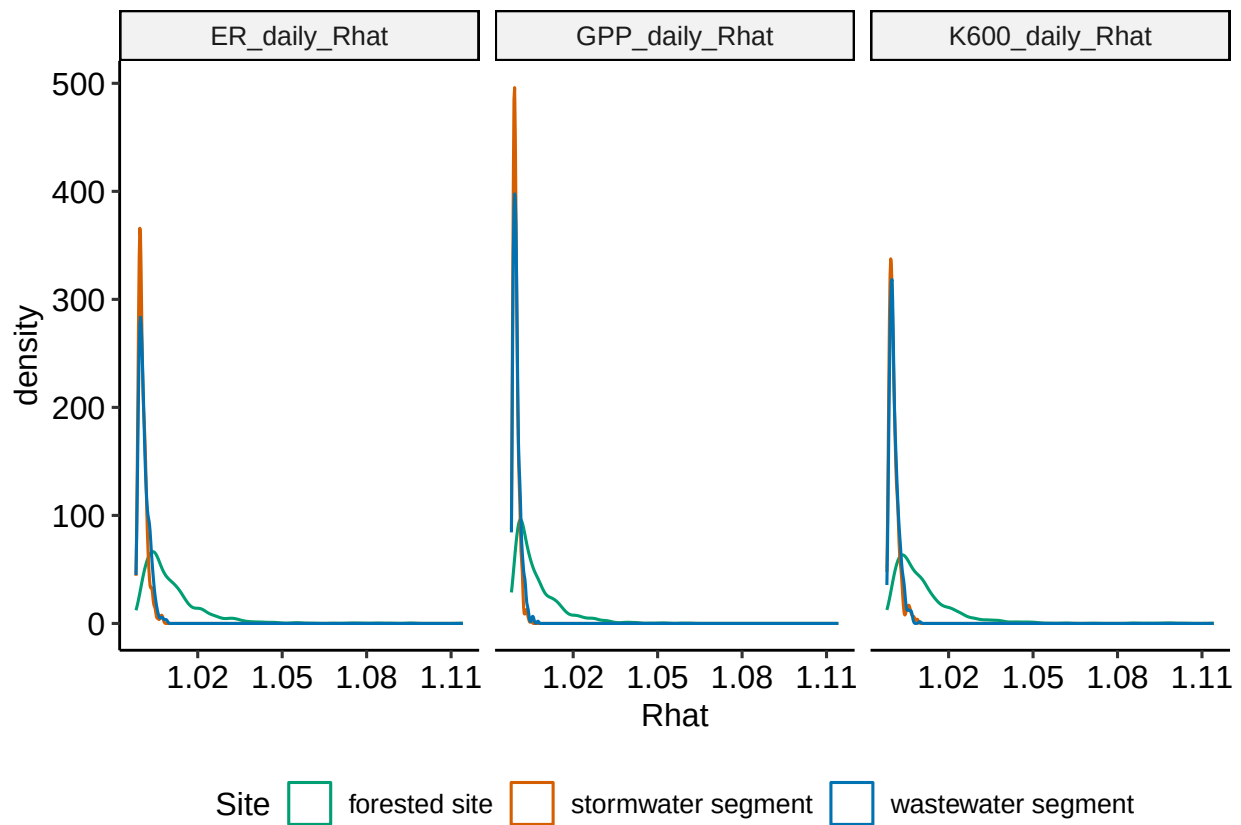
Days_Modelled	forested	stormwater dominated	wastewater dominated
Exceeds Discharge Threshold	3	5	11
GPP/ER Unrealistic	9	22	7
No Model Output	35	28	6
Passed	54	45	76

Check Model Performance

Rhat

As a first pass check, let's check the Rhats of the model outputs.

I like how Rhats are all below 1.05.



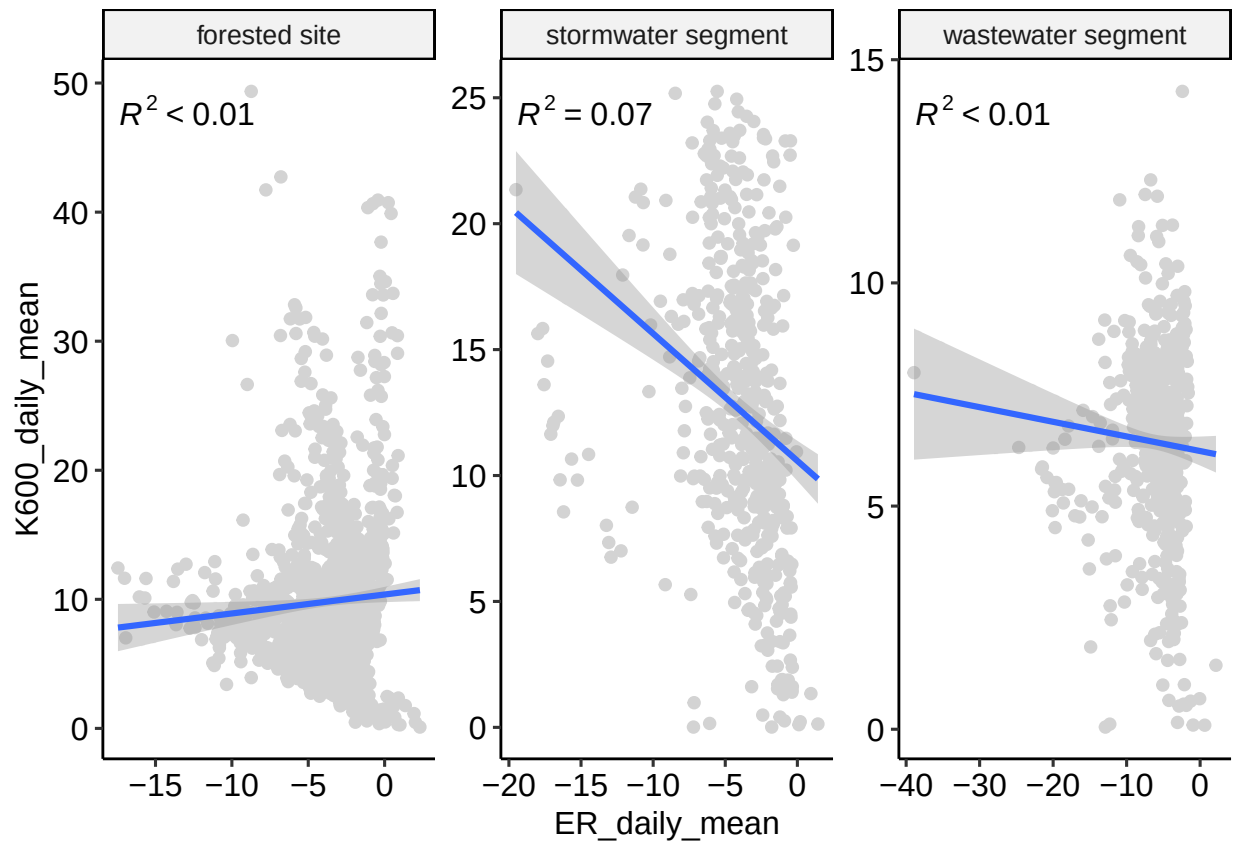
Days with rhat below 1.05 threshold.

Site	measure	Percent_of_days_below
forested	K600	98.77
forested	ER	98.77
forested	GPP	98.77
stormwater dominated	K600	100.00
stormwater dominated	ER	100.00
stormwater dominated	GPP	100.00
wastewater dominated	K600	100.00
wastewater dominated	ER	100.00
wastewater dominated	GPP	100.00

K600 vs. ER

Next, let's look at the relationship between K600 and ER.

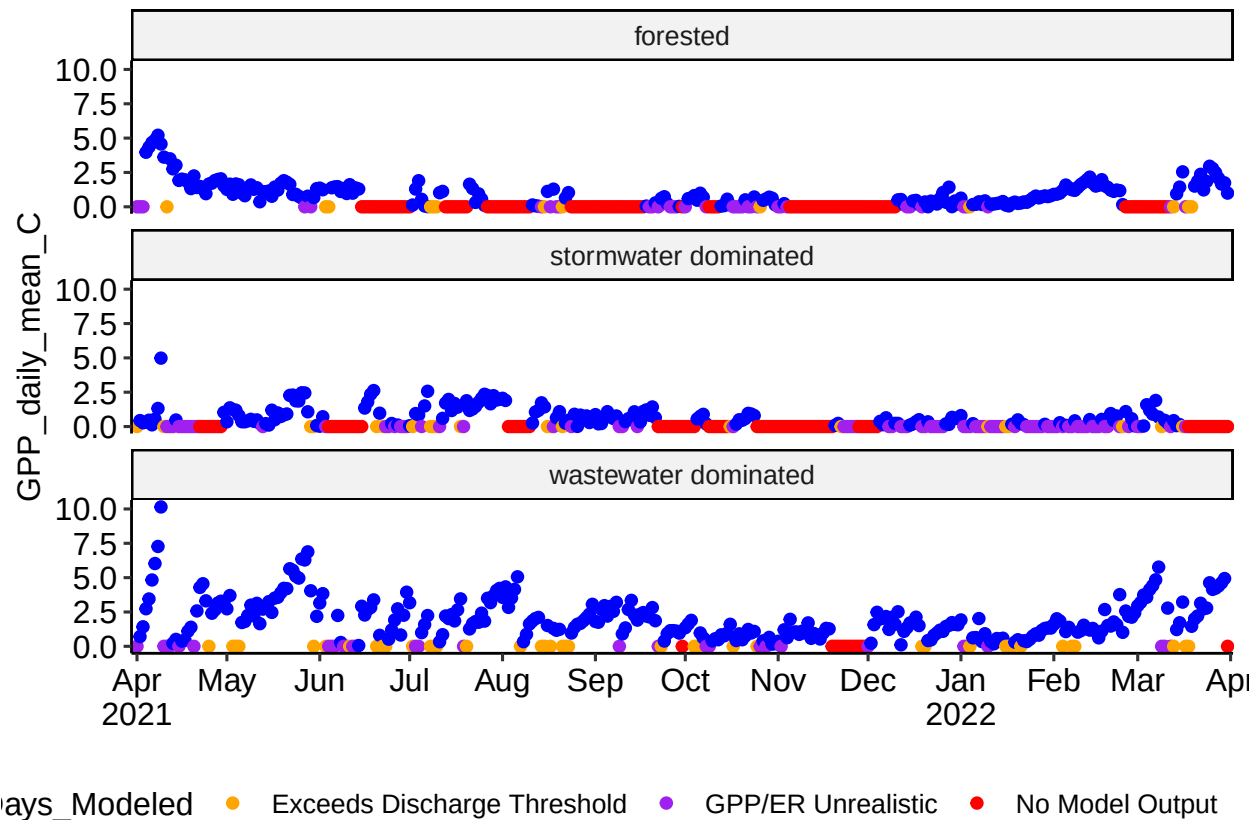
ideally, if we are not having issues with equifinality, we would expect K600 and ER to not have a clear linear relationship.



Gap Fill

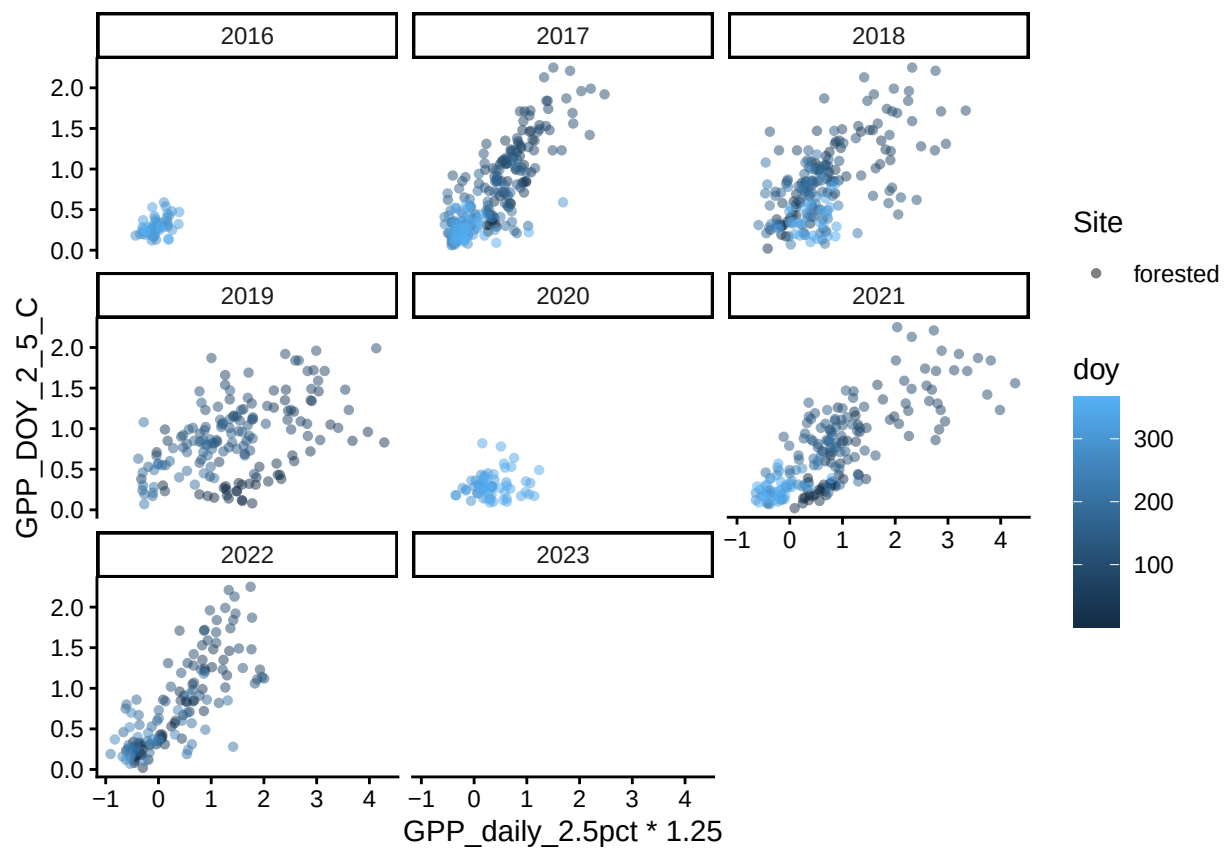
See Number of Gaps

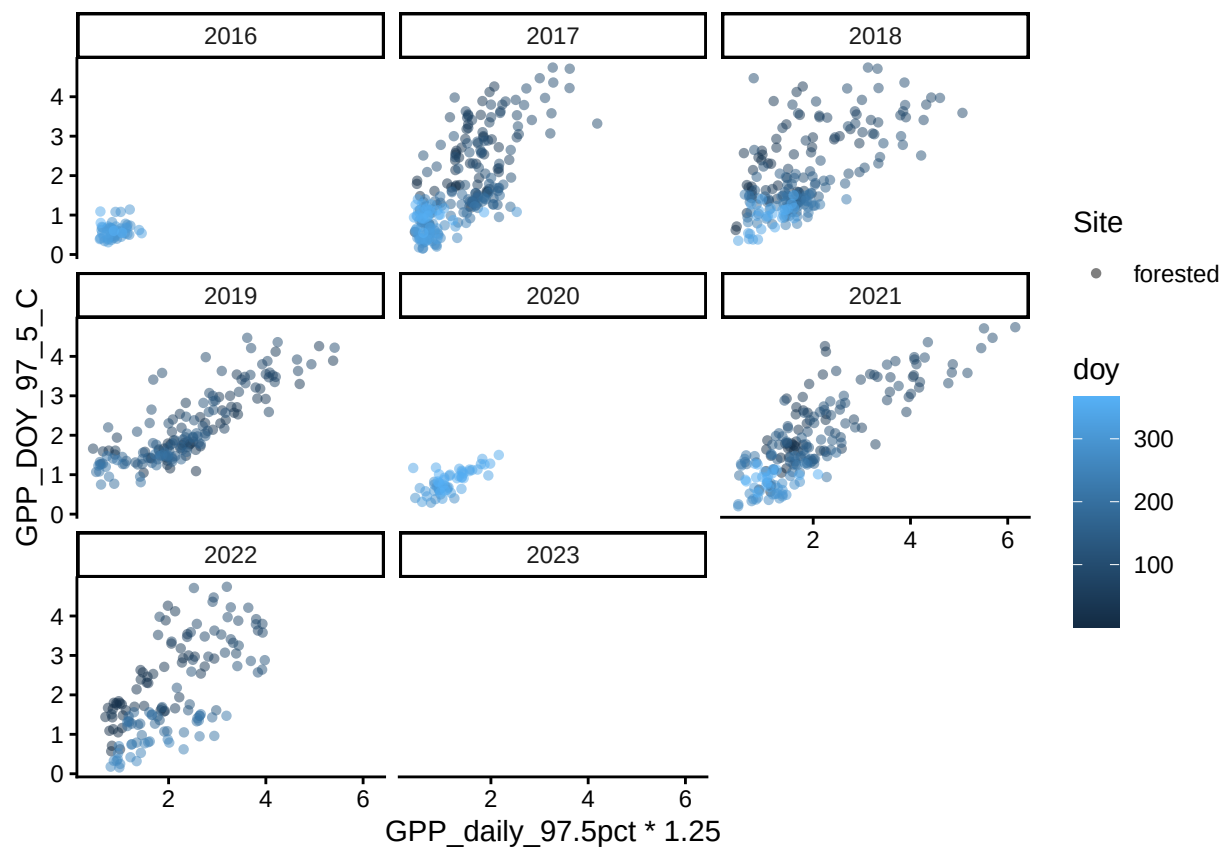
We can fix it with transmutation and other methods!

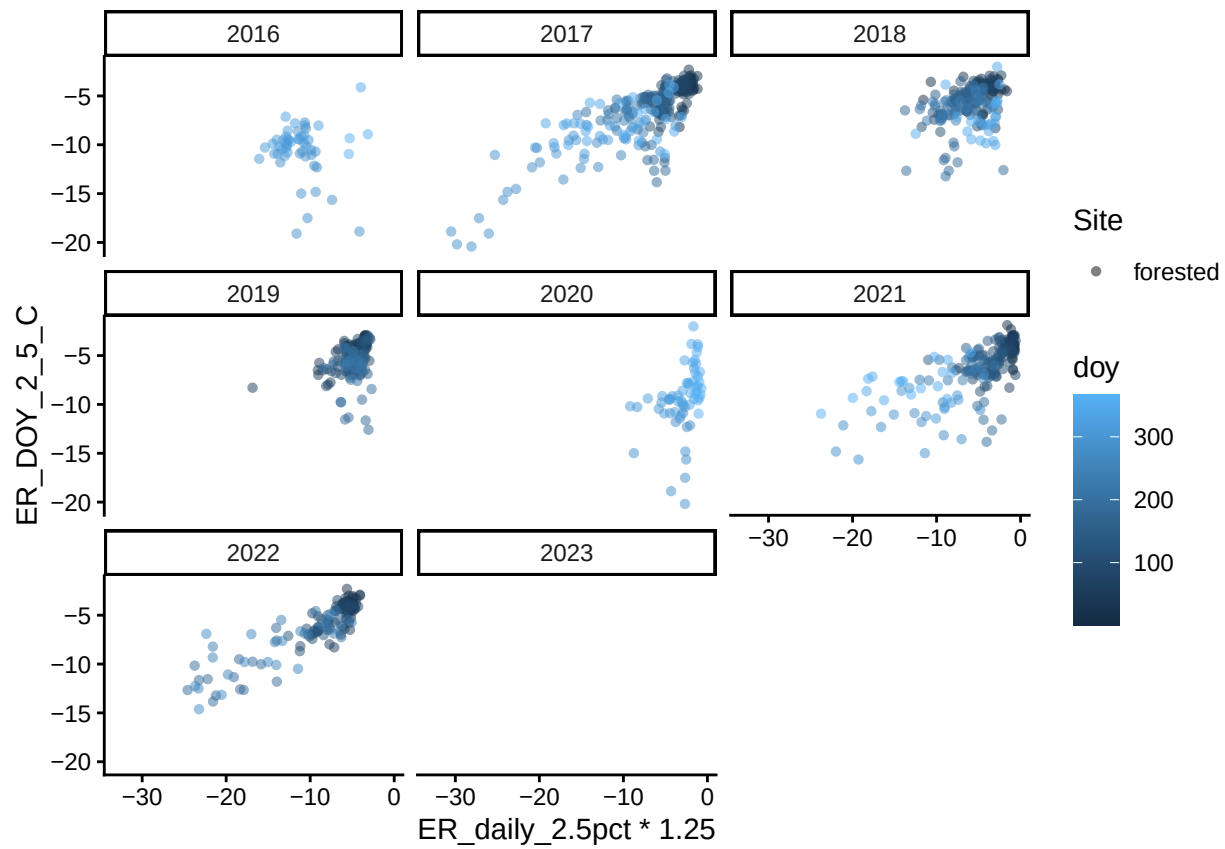


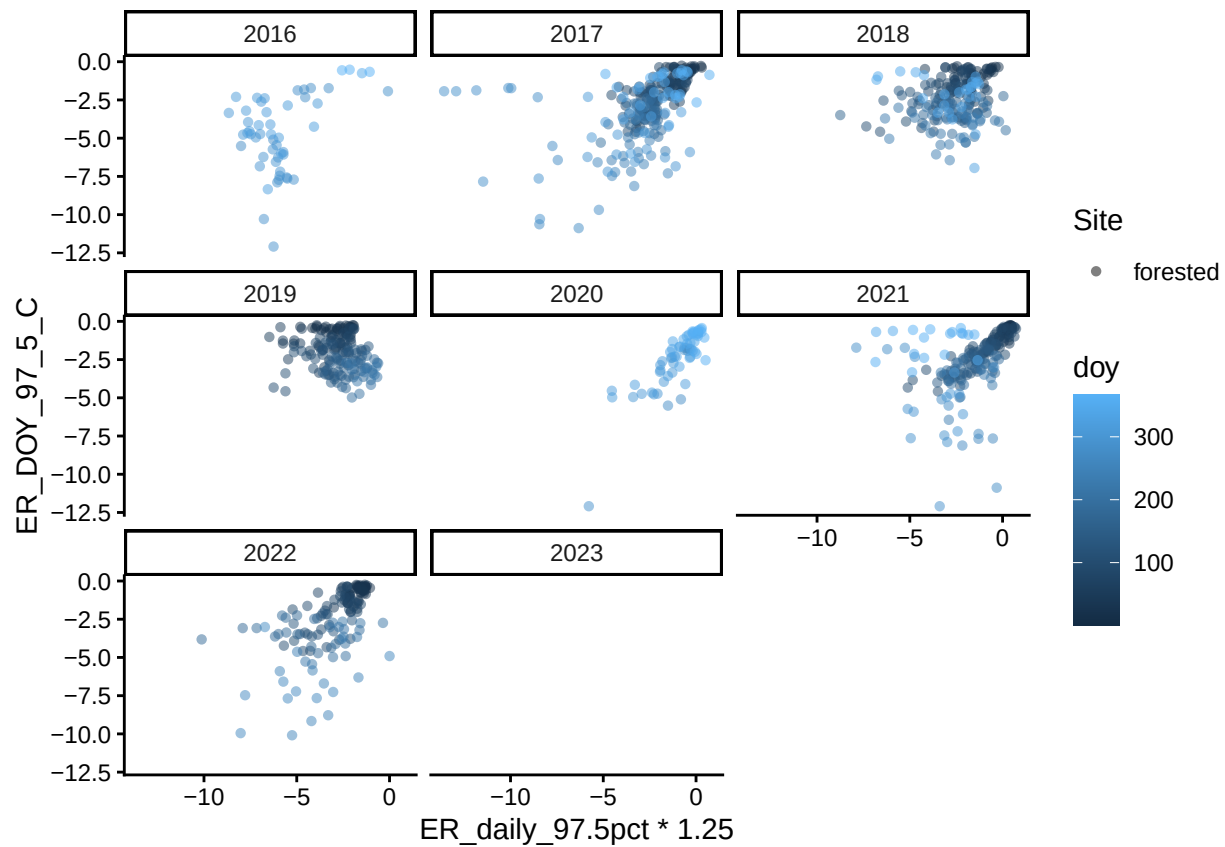
Create Quantile Range and Transmuted Values (separately)

Compare Modeled vs. Historic Quantile Range Let's see how closely the 2.5% and 97.5% quantile of the multi-year data compares to a specific day's quantile range.





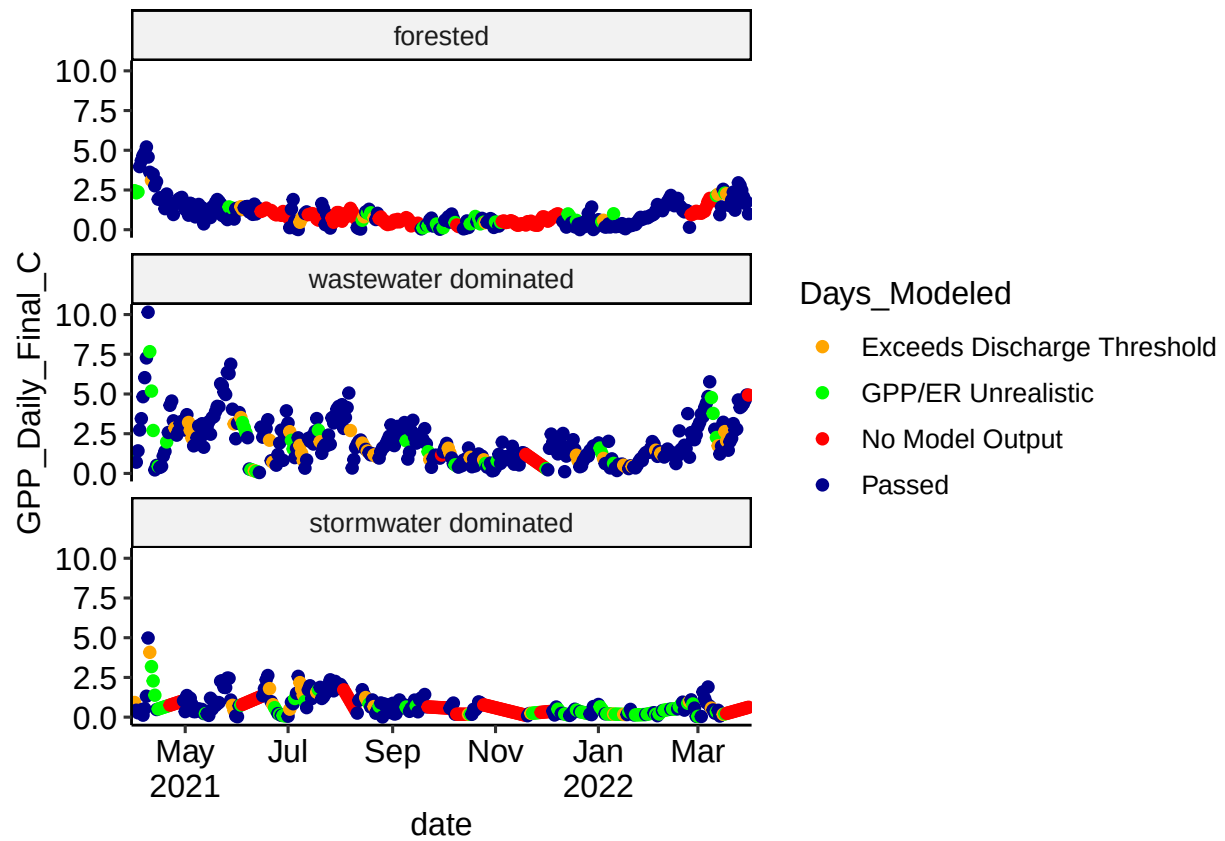




Final Data

Save the data

Here's a visualization to make it easy to see how the GPP gaps have been filled.



Model Inputs

Load Input Data

Save input data